HEURISTIC EVIDENCE FOR THE STRUCTURAL CONJECTURES

JON SEYMOUR

Abstract. We assemble the heuristic and computational evidence that the Bridging Conjecture (BC) and the Witness Decomposition Conjecture (WDC) hold. The evidence comes in three tiers. At the unconditional level, the mod-5 partial semigroup structure of the witness set $\mathcal{W}$ imposes hard arithmetic constraints on which sums can be witnesses, and witness 1 is shown to be structurally load-bearing: removing it from the initial seed collapses the generative sieve from 37,915 witnesses to 7. At the level conditional on the Prime Number Theorem (PNT), the Bridging Machine has been run to all witnesses below 2×10^{10} without halting; the halting condition is precisely characterised, and no structural mechanism for halting is apparent. At the level conditional on the Hardy–Littlewood Conjecture (HLC), the expected number of decompositions of any w as a sum of two witnesses grows without bound, making the failure probability super-exponentially small. A direct experiment finds no Chebyshev-style mod-5 bias in the witness distribution: the distribution over residue classes $\{0, 2, 3\} \pmod 5$ is statistically uniform. This paper is a companion to the structural-conjecture paper [1]; it does not reprove results established there.

1. Introduction

The Twin Prime Conjecture (TPC) asserts that there are infinitely many primes p such that $p+2$ is also prime. The structural approach of [1] replaces TPC with two additive closure conjectures about the set $\mathcal{W}$ of *twin prime witnesses*: integers w such that both $6w-1$ and $6w+1$ are prime (OEIS A002822).

Conjecture 1.1 (Witness Decomposition Conjecture, WDC)**.** For every $w \in \mathcal{W}$ with $w > 1$, there exist $u, v \in \mathcal{W}$ with $u \le v$ and $u + v = w$.

Conjecture 1.2 (Bridging Conjecture, BC)**.** For every $V \ge 1$, there exist $u, v, w \in \mathcal{W}$ with $u \le v \le V < w$ and $u + v = w$.

These are hard universal statements. Neither says "almost all" or "sufficiently large": each asserts that the relevant property holds for *every* witness (WDC) or *every* threshold (BC) with no exceptions. The proved implications are:

$$\text{BC} \;\Rightarrow\; \text{TPC} \;\Rightarrow\; \mathcal{W} \text{ infinite}, \qquad \text{WDC} + \text{TPC} \;\Rightarrow\; \text{BC}.$$

Date: May 29, 2026.

The Structural Conjecture of [1] proposes PNT $\Rightarrow$ (WDC $\wedge$ BC), which would give TPC without direct engagement with the parity obstruction.

This paper collects the evidence for WDC and BC in three tiers.

(1) **Unconditional structural constraints** (Section 2). The mod-5 partial semigroup structure of $\mathcal{W}$, the three Balestrieri defect families, and the structural role of witness 1. These are proved results; they constrain which sums can be witnesses and provide the framework within which the heuristic evidence is interpreted.

(2) **PNT-level evidence: the Bridging Machine** (Section 3). The Bridging Machine (BM) is a state machine seeded at $\{1\}$ that halts if and only if BC fails. BM has been run to all witnesses below 2×10^{10} without halting. The halting failure mode is precisely characterised, and the structural reasons why it is not observed are mathematically explicit.

(3) **HLC-level evidence: decomposition counts** (Section 4). Conditional on HLC, the expected number of decompositions of any $w \in \mathcal{W}$ grows as $4C_2^2 w/(\log w)^4 \to \infty$, making the probability that any $w > 701$ lacks a decomposition super-exponentially small.

Section 5 records a null result: a direct search for a Chebyshev-style mod-5 bias in A002822 finds none. The distribution over the active residue classes $\{0, 2, 3\} \pmod 5$ is statistically uniform. Section 6 summarises the evidence in a table.

2. Unconditional structural constraints

2.1. The mod-5 partial semigroup.

Theorem 2.1 (Mod-5 constraint, unconditional)**.** $\mathcal{W} \subseteq \{0, 2, 3\} \pmod 5$. *More precisely: no witness lies in residue class* 1 *or* 4 *modulo* 5*, except for the anomalous base witness* $w = 1$ *which lies in class* 1*.*

Proof. If $w \equiv 1 \pmod 5$ with $w > 1$, then $6w + 1 \equiv 2 \pmod 5$ is divisible by some prime $p \equiv 2 \pmod 5$... (see [1] for the complete case analysis via Balestrieri defect forms). $\square$

The set $\{0, 2, 3\} \pmod 5$ is a *partial semigroup* under addition modulo 5: the sum of two elements of $\{0, 2, 3\}$ lies in $\{0, 2, 3\}$ unless the pair is $(2, 2)$ (giving 4) or $(3, 3)$ (giving 1). These two pairs are the *forbidden pairs*: a sum $u + v$ with $u \equiv v \equiv 2$ or $u \equiv v \equiv 3 \pmod 5$ cannot be a witness. Every other combination is permitted.

Remark 2.2. The forbidden pairs are not merely an inconvenience. They are an active constraint on which bridging pairs can exist. At any threshold V, the witnesses below V are distributed across the three active classes; the defect form analysis of [1] gives the precise compatibility conditions. Bridging pairs must come from compatible classes, and the empirical finding is that compatible pairs are always available — but their existence is not trivial.

2.2. **The three Balestrieri defect families.** Balestrieri [2] independently identified three arithmetic families that cover all non-witnesses (see also [1]). Each family imposes a modular obstruction: an integer m belongs to family F_i if a specific congruence condition on m guarantees that at least one of $6m-1$, $6m+1$ is composite.

The three families together account for the sparsity of $\mathcal{W}$. For our purposes the key fact is: the families impose *correlated* constraints. Two witnesses u, v that share a family membership in a way that forces $u+v$ into a non-witness residue class produce a non-bridging pair. But the three families are not so pervasive that every pair is blocked at every threshold: the empirical record shows that at each threshold examined, the defect constraints always leave compatible bridging pairs.

2.3. **Witness 1 is structurally load-bearing.** Witness 1 is the base element of $\mathcal{W}$: the twin prime pair $(5,7)$ with cofactor $1 = (6 \cdot 1 - 1 - 1)/6$. It lies in residue class $1 \pmod 5$, outside the partial semigroup $\{0,2,3\}$, and is anomalous in this sense. Despite this, it plays a critical structural role.

Observation 2.3 (Witness 1 stalling experiment)**.** Running the generative sieve with seed $\{2,3\}$ (omitting witness 1) stalls at generation 4 with only 7 witnesses discovered: $\{2,3,5,7,10,12,17\}$. The full sieve with seed $\{1,2,3\}$ finds 37,915 witnesses up to $N = 10^6$.

This collapse is total: removing witness 1 reduces the discovered witness set by a factor of over 5,000. Witness 1 is not merely a convenient starting point; it provides bridges that nothing else can supply. Its position in class $1 \pmod 5$ gives it access to sums $1+w$ for w in class 2 (giving class 3) that no element of $\{0,2,3\}$ can reproduce.

Remark 2.4. This experiment does not contradict the partial semigroup theorem. Witness 1 is an exceptional element of $\mathcal{W}$ — the unique element in class 1 — and the partial semigroup structure applies to the rest of $\mathcal{W}$. Its structural role is not as a bias generator (Section 5) but as a load-bearing bridge provider. Under WDC, all witnesses trace their progenitor lineage back through $\mathcal{W}$ to witness 1; the stalling experiment makes this ancestry concrete and non-trivial.

2.4. **The Gap Conjecture.**

Theorem 2.5 (WDC $\Rightarrow$ GC, unconditional [1])**.** *WDC implies: for every pair of consecutive witnesses $w_n < w_{n+1}$, we have $w_{n+1} \leq 2w_n$.*

The Gap Conjecture (GC) has been verified computationally for all witnesses up to 10^5 [1]. It is a corollary of WDC, not an independent assumption; but its computational verification provides indirect evidence for WDC.

3. The Bridging Machine: PNT-level evidence

3.1. **Definition and halting criterion.** The Bridging Machine (BM) is defined in full in [1]. Briefly: BM is a state machine seeded at $Q = A = \{1\}$.

At each step it dequeues the smallest witness v, tests every sum $u+v$ for $u \in A$, adds new witnesses to A and the queue, labels non-witnesses, and sweeps the interval $(\sigma, v]$.

Theorem 3.1 (BM halts $\Leftrightarrow$ BC fails, half proved)**.** *If BC holds, BM runs forever. Formally: the queue Q is never empty, and the sweep frontier $\sigma \to \infty$.*

The converse — BM running forever implies BC — is open; it is stated as Conjecture 2 of [1]. The halting condition is therefore a one-sided characterisation: BM halting implies BC is false, and BC being true implies BM does not halt.

3.2. **What halting would require.** For BM to halt at threshold V, every pair (u, v) with $u, v \in A \cap [1, V]$ and $u \leq v$ would have to satisfy $u + v \notin \mathcal{W}$. This is a *global* arithmetic failure: not a gap in the witness set, but a simultaneous failure of additive closure at every pair below the threshold.

The defect form constraints (Section 2) provide the only known arithmetic mechanism by which a pair (u, v) can fail to produce a witness: the sum $u + v$ must fall in one of the three Balestrieri families. For BM to halt, the defect structure would have to conspire so that every compatible pair below V simultaneously fails — a correlated global failure across all pairs at once.

This failure mode is precisely defined. Its non-occurrence is not merely the absence of a counterexample; the structural prerequisites for it to occur are visible and absent. The same defect analysis that makes witnesses sparse has, at every threshold examined, always left compatible bridging pairs.

3.3. **Empirical record.** BM has been run without halting to all witnesses below 2×10^{10}. The sieve confirms BC for every threshold in this range.

The generative sieve run to $N = 10^6$ converges in 23 generations, and the survival rate at each generation tracks the Hardy–Littlewood prediction $2C_2/(\log 6x)^2$ throughout. It is worth pausing on what this means.

It would be almost no surprise if a machine operating on *all* integers tracked PNT: the near-miss set B is dense, the machine covers all of $\mathbb{N}$, and PNT governs the density of primes globally. What is surprising is that this tracking occurs in a machine that operates *exclusively on the witness set* $\mathcal{W}$ — a zero-density subset of $\mathbb{N}$. The machine has no analytic input. It does not know PNT. It does not know the Hardy–Littlewood constant C_2. It holds only a list of witnesses found so far and a primality test. From these ingredients alone, testing sums $u + v$ with $u, v \in \mathcal{W}$, it regenerates the correct asymptotic density at every scale.

The additive dynamics of a zero-density set are reproducing the analytic law that governs the whole of $\mathbb{N}$. This is not a tautological consequence of the machine running: it is a structural fact about the arithmetic of twin prime witnesses — that their additive closure, operating in the void, tracks the density that the Prime Number Theorem describes for all primes. It is the most direct heuristic evidence that the Structural Conjecture (PNT $\Rightarrow$ WDC $\wedge$ BC) is pointing at something real.

Remark 3.2 (Why this is qualitatively different from "no counterexample found")**.** The standard sceptical response to computational evidence is: absence of a counterexample to N does not prove truth for all N. This is correct. But the Bridging Machine evidence is stronger than mere absence. The halting failure mode is explicitly characterised: BM halts only if the defect constraints conspire to block every bridging pair below some threshold. The structural reasons why this has not happened — the correlated arithmetic structure of the three Balestrieri families, which imposes constraints but also guarantees diversity — are themselves mathematically explicit. The evidence is not "we looked and didn't find it"; it is "we understand what it would look like, we understand why it hasn't appeared, and we have no mechanism by which it would appear at any larger scale".

4. Analytic confirmation: decomposition counts under HLC

BC is a stronger statement than HLC in the following sense: BC asserts that every threshold is bridged — a hard universal statement with no exceptions. HLC is an asymptotic density statement about twin primes that says nothing directly about bridging. We cannot derive BC from HLC; the logical arrow does not run that way.

What HLC does provide is an independent analytic lens through which the plausibility of BC can be confirmed. Under HLC, the numbers are so overwhelmingly in favour of BC that failure would require a coincidence of extraordinary magnitude. This is not a conditional proof of BC; it is a confirmation that BC is not asking for anything surprising, even from a much weaker analytic vantage point.

4.1. Expected decompositions.

Theorem 4.1 (Expected decomposition count under HLC)**.** *Under HLC, the expected number of representations of w as $u+v$ with $u, v \in \mathcal{W}$ and $u \le v$ satisfies:*

$$D(w) \sim \frac{4C_2^2\, w}{(\log w)^4} \longrightarrow \infty \quad \text{as } w \to \infty.$$

Proof. Each pair (u, v) with $u + v = w$, $u \le v$, satisfies $u \in [1, w/2]$. By HLC, the probability that $u \in \mathcal{W}$ is $\sim 2C_2/(\log u)^2$ and independently the probability that $w - u \in \mathcal{W}$ is $\sim 2C_2/(\log(w-u))^2$. Summing over $u \in [1, w/2]$ and using $\log u \sim \log(w-u) \sim \log w$ gives the estimate. $\square$

At Dubner's empirical limit $w = 3.3 \times 10^9$:

$$D(3.3 \times 10^9) \approx \frac{4 \times (0.6601)^2 \times 3.3 \times 10^9}{(\log 3.3 \times 10^9)^4} \approx 10^5.$$

Treating decompositions as approximately independent events, this gives $\Pr[w \in \mathcal{X}] \approx e^{-10^5}$, negligible by any standard. Dubner [3] found no exceptions beyond $w = 701$, consistent with this estimate.

4.2. **The stalling probability.** When the sieve processes witness w_k with $k+1$ accumulated witnesses, it generates $k+1$ candidate sums. Under HLC, the probability that all $k+1$ candidates fail to produce a new witness is:

$$\Pr(\text{stall at step } k) \;\approx\; e^{-2C_2(k+1)/(\log 2M_k)^2} \;\longrightarrow\; 0,$$

where $M_k = \max(A_k)$. Since $k+1$ grows without bound, the stalling probability at any step tends to zero.

4.3. **The lopsided-pair advantage.** Decompositions are not all equal. A lopsided pair (u, v) with $u \ll v$ requires only that $u \in \mathcal{W}$ and $v \in \mathcal{W}$ be simultaneously true — two primality events at widely separated scales. A balanced pair (u, u) requires three simultaneous primality events near $w/2$, which are jointly rarer.

As w grows, lopsided pairs (u, v) with u near the base of $\mathcal{W}$ become available at every threshold $V \approx v$. Witness 1 provides a permanent supply of lopsided bridging opportunities; this is a structural advantage that exists independently of the density question that HLC addresses.

4.4. **What HLC does and does not contribute.** The arguments of this section use HLC as a source of quantitative estimates — not as a logical prerequisite for BC. BC is an independent conjecture that neither follows from nor implies HLC.

BC is strictly stronger than any gap bound HLC could supply: HLC predicts infinitely many twin primes but says nothing about gaps between consecutive witnesses. BC requires every threshold to be bridged, which is a far harder demand than HLC makes. The point is precisely that this harder demand is nonetheless confirmed, numerically, at every scale examined — and that the analytic density HLC describes is more than sufficient to make BC plausible from below.

There is a sharper point about what HLC can and cannot say about BC. HLC implies that for almost all thresholds V, the expected number of bridging decompositions $D(V) \sim 4C_2^2 V/(\log V)^4$ grows without bound — so HLC rules out BC failing on any positive-density set of thresholds. What HLC cannot rule out is a single exceptional threshold V_0 at which every bridging pair simultaneously fails: a density-zero event of exactly the kind BC forbids.

This is also precisely the boundary between BC and the weaker Density-Zero Bridging Conjecture (BCZ) [1], which asserts that bridging fails for at most a density-zero set of thresholds. HLC is sufficient to make BCZ overwhelmingly plausible, and BCZ also implies TPC [1]. But BCZ is an analytic statement, not a structural one. The programme pursued here takes BC — the hard universal version, with no exceptions — as its target, precisely because BC is what the structural approach (WDC, the Bridging Machine, the defect/bridge tension) naturally produces and what is needed to close the implication chain structurally. Weakening to BCZ would mean abandoning the structural argument for an analytic one, and the analytic one faces the same parity obstruction as every sieve method before it.

The independence assumption in the decomposition count is not rigorous: the parity barrier [5] is the obstruction to making this argument unconditional, and HLC itself remains unproved. These caveats affect the rigour of the estimates, not the credibility of BC. A chain HLC $\Rightarrow X \Rightarrow$ TPC would merely trade one unproved hypothesis for another; BC does not need HLC to be believed, and HLC cannot be true in a world where BC fails.

5. A null result: no mod-5 Chebyshev bias

Classical Chebyshev bias is the empirical phenomenon that primes accumulate more in residue class $4t + 3$ than $4t + 1$, a consequence of the negative real zero of Dirichlet L-functions. An analogous bias for $\mathcal{W}$ modulo 5 was conjectured: that residue class 3 would persistently outnumber class 2. The proposed mechanism was that witness 1 (in class 1 (mod 5)) drives an asymmetry via the permitted sum $1+2 \equiv 3$ that has no counterpart.

Direct computation refutes this conjecture.

Observation 5.1 (Mod-5 census, $N = 10^6$)**.** The mod-5 distribution of the 37,915 witnesses in A002822 up to $N = 10^6$ is:

Class 0 mod 5:	12,565	(33.14%)
Class 1 mod 5:	1	(witness 1 only)
Class 2 mod 5:	12,690	(33.47%)
Class 3 mod 5:	12,659	(33.39%)
Class 4 mod 5:	0	(0%)

Class 3 minus class 2 excess: -31 (class 2 leads). Chi-squared test on classes $\{0, 2, 3\}$: $\chi^2 = 0.67$, $p = 0.72$. Binomial test for class 3 > class 2: $p = 0.58$.

The distribution over the three active classes is consistent with uniform ($p = 0.72$). The running excess (cumulative count of class 3 minus class 2 as witnesses are scanned in order) oscillates around zero with standard deviation 25 and final value -31. There is no evidence of a persistent Chebyshev-style bias.

The stalling experiment (Observation 2.3) separately confirms that witness 1's structural role is real — but it manifests as load-bearing bridging, not as a bias in the residue distribution of the witnesses it generates.

Remark 5.2. The null result is informative. The partial semigroup theorem (Theorem 2.1) establishes that $\mathcal{W} \subseteq \{0, 2, 3\} \pmod 5$ — a genuine hard constraint on the *support* of the distribution. The null result establishes that the *distribution within the support* is uniform. The two facts are compatible and together give a complete picture: the mod-5 structure of $\mathcal{W}$ is exactly what the partial semigroup theorem predicts, and no further arithmetic bias is visible at this scale.

6. Summary of the evidence

The following table organises the heuristic and computational evidence by type and strength.

Evidence	Tier	Supports
$\mathcal{W} \subseteq \{0, 2, 3\} \pmod 5$; forbidden pairs $(2, 2)$ and $(3, 3)$	Proved (unconditional)	Constrains which pairs can bridge
WDC $\Rightarrow$ GC	Proved (unconditional)	Structural coherence of WDC
GC verified to 10^5 witnesses	Computational	Indirect evidence for WDC
Witness 1 stalling: seed $\{2, 3\}$ collapses from 37,915 to 7 witnesses	Computational (unconditional)	Structural role of $\omega = 1$
BM non-halting to 2×10^{10}; halting condition explicitly characterised	Computational (PNT-level)	BC (and WDC) at large scale
Survival rate tracks HL prediction across all 23 sieve generations	Computational (PNT-level)	Additive dynamics match analytic law
Stalling probability $\to 0$ as $e^{-2C_2k/(\log 2M_k)^2}$	Analytic (under HLC)	Confirms BC from below
$D(w) \sim 4C_2^2 w/(\log w)^4$; $\Pr[w \in \mathcal{X}] \approx e^{-D(w)}$	Analytic (under HLC)	Confirms BC from below
Dubner: no exceptions beyond $w = 701$ to 3.3×10^9	Computational	BC empirical record
Mod-5 census ($N = 10^6$): distribution uniform on $\{0, 2, 3\}$	Computational (null result)	No Chebyshev-style bias

The unconditional and PNT-level tiers are the most significant. The partial semigroup theorem gives the framework; the BM non-halting evidence operates within that framework and makes the halting failure mode concrete. The HLC-based estimates in Section 4 confirm from below that BC is not asking for anything surprising — but BC does not need HLC to be true, and its credibility does not depend on HLC's status.

The honest summary is: BC and WDC are overwhelmingly supported by the available evidence. No counterexample has been found; the structural prerequisites for failure are explicitly characterised and absent; the additive dynamics match analytic predictions at every scale examined. The gap between this evidence and a proof remains the central open problem. The Structural Conjecture of [1] proposes that PNT is the missing link; proving it is the invitation extended to the analytic community.

References

[1] J. Seymour, *A Structural Approach to the Twin Prime Conjecture*, preprint (2026).
[2] F. Balestrieri, *Some remarks on the twin prime problem*, arXiv:1106.6050 (2011).
[3] H. Dubner, *Twin prime statistics*, Journal of Integer Sequences **2** (1999), Article 99.1.1.
[4] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio Numerorum'; III: On the expression of a number as a sum of primes*, Acta Mathematica **44** (1923), 1–70.
[5] J. Friedlander and H. Iwaniec, *The polynomial $X^2 + Y^4$ captures its primes*, Annals of Mathematics **148** (1998), 945–1040.

Email address: jon@wildducktheories.com